

Erez A. Zimmerman

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Research Profile

I study supernovae to constrain their progenitor stars, pre-explosion mass loss, and explosion mechanisms. In particular, I have studied nearby supernovae, including the decadal supernova **2023ixf** (Zimmerman et al. 2024, published in *Nature*), for which I obtained first-of-its-kind Ultraviolet spectroscopy using the *Hubble Space Telescope*, and the extremely faint Iax supernova **2024vjm** (Zimmerman et al., accepted in principle at *Nature Astronomy*), for which I analyzed a pre-explosion *Euclid* space telescope image containing the explosion site. I have led several successful telescope proposals for facilities such as the **Hubble Space Telescope**, **Gemini**, and the **VLT**.

Research Leadership and Collaborations

- I am a member of major transient collaborations including **BlackGEM**, **LAST**, **ePESSTO+**, **SoXS**, and formerly **ZTF**.
- I lead the BlackGEM consortium bi-weekly transient meetings
- I developed **CAST**, a TOM-based transient marshal for the Weizmann Observatory's LAST collaboration.

Education

Weizmann Institute of Science	Rehovot, Israel
Ph.D. in Astrophysics; advisor: Prof. Avishay Gal-Yam	2021 – expected Oct. 2026
Weizmann Institute of Science	Rehovot, Israel
M.Sc. in Astrophysics; advisor: Prof. Avishay Gal-Yam	2018 – 2021
Technion – Israel Institute of Technology	Haifa, Israel
B.Sc. in Physics	2015 – 2018

Honors and Awards

Weizmann School of Science Prof. Israel Dostrovsky Prize of Excellence	2026
Weizmann School of Science Haim Holtzman Memorial Prize	2025

First-Author Publications

The Complex Circumstellar Environment of Supernova 2023ixf

Erez A. Zimmerman, Ido Irani, Ping Chen, Avishay Gal-Yam et al.

Nature, 627, 759–762, 2024

A Faint Progenitor System for the Faint Supernova 2024vjm

Erez A. Zimmerman, Avishay Gal-Yam et al.

Nature Astronomy, accepted in principle

Co-Author Publications

The Type II In SN 2025cbj coincidence with the high-energy neutrino IceCube-250421A

Simone Garrappa, Erez A. Zimmerman, et al.

[A&A, 708, A223 \(2026\)](#)

The flash-ionised SN Ibn 2025kzr: H-free circumstellar material formed during a precursor outburst 55 days prior to explosion

S. de Wet, G. Leloudas, D. A. H. Buckley, N. Erasmus, P. J. Groot, Erez A. Zimmerman, et al.

[A&A, accepted \(2026\)](#)

SN 2019vxm: A Shocking Coincidence between Fermi and TESS

Zachary G. Lane et al. including Erez A. Zimmerman

[ApJ, 1003, 19 \(2026\)](#)

The Large Array Survey Telescope Pipeline: II. Image subtraction and transient detection

R. Konno et al. including Erez A. Zimmerman

[A&A, 710, A406 \(2026\)](#)

XSNAP: An X-ray Supernova Analysis Pipeline with Application to the Type II Supernova 2024ggi

Ferdinand, Wynn V. Jacobson-Galán, Mansi M. Kasliwal, Erez A. Zimmerman

[ApJ, 1001, 26 \(2026\)](#)

Extremely Stripped Supernova Reveals a Silicon and Sulfur Formation Site

Steve Schulze et al. including Erez A. Zimmerman

[Nature, 644, 634–639 \(2025\)](#)

A low mass, binary-stripped envelope for the Type IIb SN 2024abfo

S. de Wet, G. Leloudas, D. A. H. Buckley, N. Erasmus, P. J. Groot, Erez A. Zimmerman

[A&A, 704, A89 \(2025\)](#)

A Search for Minute-timescale Flares from the Transient AT 2024wpp

Eran O. Ofek et al. including Erez A. Zimmerman

[ApJ, 993, 76 \(2025\)](#)

Low-luminosity Type IIP Supernovae from the Zwicky Transient Facility Census of the Local Universe. I. Luminosity Function, Volumetric Rate

Kaustav K. Das et al. including Erez A. Zimmerman

[PASP, 137, 044203 \(2025\)](#)

The Progenitor Star of SN 2023ixf: A Massive Red Supergiant with Enhanced, Episodic Pre-Supernova Mass Loss

Yu-Jing Qin, Keming Zhang, Joshua Bloom, Jesper Sollerman, Erez A. Zimmerman, et al.

[MNRAS, 534, Issue 1, 271–280 \(2024\)](#)

The BlackGEM Telescope Array. I. Overview

Paul J. Groot et al. including **Erez A. Zimmerman**

PASP, 136, 115003 (2024)

The Early Ultraviolet Light Curves of Type II Supernovae and the Radii of Their Progenitor Stars

Ido Irani et al. including **Erez A. Zimmerman**

ApJ, 970, 96 (2024)

SN 2022oqm—A Ca-rich Explosion of a Compact Progenitor Embedded in C/O Circumstellar Material

Ido Irani et al. including **Erez A. Zimmerman**

ApJ, 962, 109 (2024)

The Prevalence and Influence of Circumstellar Material Around Hydrogen-rich Supernova Progenitors

Rachel J. Bruch, Avishay Gal-Yam, Ofer Yaron, Ping Chen, Nora L. Strotjohann, Ido Irani, **Erez Zimmerman**, et al.

ApJ, 952, 119 (2023)

A WC/WO Star Exploding Within an Expanding Carbon–Oxygen–Neon Nebula

Avishay Gal-Yam et al. including **Erez A. Zimmerman**

Nature, 601, 201–204 (2022)

Candidate Tidal Disruption Event AT2019fdr Coincident with a High-Energy Neutrino

Simeon Reusch et al. including **Erez Zimmerman**

Phys. Rev. Lett. 128, 221101 (2022)

The Spectacular Ultraviolet Flash from the Peculiar Type Ia Supernova 2019yvq

A. A. Miller, M. R. Magee, A. Polin, K. Maguire, **E. Zimmerman** et al.

ApJ, 898, 56 (2020)

Accepted Telescope Time as PI

Infant Supernovae: Observing Massive Stars as They Explode with the Large Array Survey Telescope

Gemini 2025B

4.5 hours

Observing the Local Environment of the Nearby Faintest Iax SN 2024vjm with MUSE Narrow Field Mode

ESO Period 114 DDT

2.9 hours, VLT/MUSE

Observing the Circumstellar Interaction of the 7 Mpc SN 2024ggi

30 ks

Chandra DDT 25508981

High-Resolution Spectroscopy of One of the Nearest Supernovae in a Decade: SN 2024ggi ESO Period 113

DDT

2.7 hours, VLT/UVES

Explosions in Real Time: Rapid UV Supernova Flash Spectroscopy

22 orbits

HST Cycle 30

Infant Supernovae: Observing Massive Stars as They Explode

OPTICON 2024B

Liverpool Telescope 5 hr + TNG 0.3 nights

Infant Supernovae: Observing Massive Stars as They Explode from the Ground and Space

Gemini 2023A
0.9 nights

Explosions in Real Time: Rapid UV Supernova Flash Spectroscopy

OPTICON 2023A

TNG 1.0 night

Observational Experience

- **Hubble Space Telescope:** 22 orbits.
- **VLT/MUSE:** 1 remote observing run.
- **VLT/UVES:** 4 remote observing runs.
- **NTT, La Silla Observatory, Chile:** 9.5 nights + ToO observations in ePESSTO+
- **Lowell Discovery Telescope, Lowell Observatory, Arizona, USA:** 1 night as well as ToO observations.
- **Neil Gehrels Swift Observatory:** 40+ approved ToO observations
- **Aristarchos Telescope, Chelmos Observatory, Greece:** 6 nights.
- **Target-of-opportunity activations:** extensive experience triggering rapid ToO photometry and spectroscopy with multiple facilities and instruments.

Teaching

Lecturer, Open University of Israel

2022 – 2026

Secrets of the Universe: A Window to Astronomy

Instructor for a general astronomy course covering the history of astronomy, the Solar System, astronomical instrumentation, stellar physics, supernovae and supernova remnants, galaxy formation, and cosmology. Average student rating: 4.3/5.

Invited Talks

SN 2023ixf Early Emission and Circumstellar-Material Mapping

Jun. 2024

Review talk, SN 2023ixf: The Decadal Supernova in M101 Workshop, European Southern Observatory, Garching, Germany.

Ultraviolet Observations Resolve the Explosion of Supernova 2023ixf in Messier 101 within Its Complex Circumstellar Environment

Jan. 2024

American Astronomical Society Meeting 243, New Orleans, LA, USA.

Invited Seminars

Dying Stars Shed Light on Late-Stage Mass Loss

Nov. 2025

Invited seminar, Technion, Haifa, Israel.

A Framework to Map Circumstellar Material Around Core-Collapse Supernovae

May 2025

Invited seminar, Hebrew University of Jerusalem, Jerusalem, Israel.

The Complex Circumstellar Environment of Supernova 2023ixf Feb. 2024
Tel Aviv University Student Seminar, Tel Aviv University, Tel Aviv, Israel.

Selected Contributed Talks

Dying Stars Shed Light on Late-Stage Mass Loss Oct. 2026
Type II Supernovae: The Chaotic Deaths of Supergiants and their Cosmic Implications, San Pedro de Atacama, Chile.

A Faint Progenitor System for the Faint Supernova 2024vjm Jun. 2026
American Astronomical Society Meeting 248, Pasadena, CA, USA.

Dying Stars Shed Light on Late-Stage Mass Loss Nov. 2025
Stellar Populations and Their Explosions: Bridging the Gap Workshop, Max Planck Institute for Astrophysics, Ringberg Castle, Germany.

SN 2024vjm: A Faint Supernova and a Fainter Companion Sep. 2025
BlackGEM Consortium Fall Meeting 2025, Hamburg Observatory, Hamburg, Germany.

A Framework to Map the Circumstellar Material Around Red Supergiant Supernovae Jul. 2025
Israel Physical Society Meeting, Technion, Haifa, Israel.

Ultraviolet Observations of Infant Supernovae Mar. 2025
Transients from Space, STScI, Baltimore, MD, USA.

SN 2024ggi and Infant Supernovae: Early Light Curve and Circumstellar Material Sep. 2024
BlackGEM Consortium Fall Meeting 2024, University of Manchester, Manchester, UK.

Early and UV Observations of SN 2023ixf, a Core-Collapse Supernova in the Nearby M101 Galaxy Aug. 2023
SNeX – SuperNova EXplosions Conference, Technion, Haifa, Israel.

High-Velocity Circumstellar Wind Around the Type II Supernova SN 2020pni Jun. 2022
Zwicky Transient Facility Spring Meeting 2022, CNRS/IN2P3, Paris, France.

Posters

A Faint Progenitor System for the Faint Supernova 2024vjm May 2026
IAU Symposium 406 – Future Landscape of Astrophysical Transients, Turku, Finland.

The Complex Circumstellar Environment of Supernova 2023ixf Aug. 2024
International Astronomical Union General Assembly, Cape Town, South Africa, remote.

High-Velocity Circumstellar Wind Around the Type II Supernova SN 2020pni Feb. 2022
Israel Physical Society 67th Meeting, Ben-Gurion University, Be'er Sheva, Israel.

Outreach, Service, and Media

- I peer-review manuscripts for the Astrophysical Journal (ApJ) and Astronomy & Astrophysics (A&A).
- Weizmann mission to LATAM, meeting with friends of Weizmann in Chile and Brazil August 2026
- Recorded lecture for Israeli Space Week, *Telescopes in Space Reveal the Mysteries of the Universe*, for ages 13–14, Hebrew Jan. 2025
- Guest lecture for the Israeli Astronomical Society, *We Are All Stardust: Supernovae and the Origin of the Elements*, Hebrew Sep. 2024
- Guest, *Voice of the Universe* podcast, episode 1, Hebrew Apr. 2024
- Guest, BBC *Science in Action*, “Brightest Supernova in a Decade” May 2023
- Press coverage of Zimmerman et al. 2024 in [Weizmann](#), [Stockholm University](#), [Keck Observatory](#), [Ynet](#), [Mako](#), and Kan Radio.

Other Professional Experience and Affiliations

Weizmann–Helmholtz Multi-Messenger School Fellow	2021 – 2026
Summer Student at the European Organization for Nuclear Research (CERN)	Jun. 2018 – Sep. 2018
Secretary General of the Technion student union	2017
Head of Physics Student council, Technion	2016-2017
Mandatory Military Service	2012-2015